

Lab Exercise 2: Preprocessing the Dataset Using Normalization Techniques

Objective

To apply normalization techniques on an Education and Career Success dataset and analyze the impact of normalization on different numerical attributes. The study focuses on preparing the dataset for machine learning models by scaling features to comparable ranges.

Part A: Dataset Understanding

1) Identify the domain of the dataset.

The dataset belongs to the **Education Analytics and Career Success Analysis** domain. It contains information related to students' academic performance, practical experience, skills, and career outcomes.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

df = pd.read_csv('education_career_success.csv')
print(df.head(3))
```

Student_ID	Age	Gender	High_School_GPA	SAT_Score	University_GPA
0	22	Male	3.8	1450	3.6
1	24	Female	3.6	1380	3.4
2	21	Male	3.9	1520	3.8

Field_of_Study	Internships_Completed	Projects_Completed
0 Computer Science	3	7
1 Business	2	5
2 Engineering	4	9

Certifications	Soft_Skills_Score	Networking_Score	Job_Offers
0 2	8	7	3
1 3	7	6	2
2 4	9	8	4

Starting_Salary	Career_Satisfaction	Years_to_Promotion	Current_Job_Level
0 85000	8	2	Mid
1 65000	7	3	Mid
2 120000	9	1	Senior

Work_Life_Balance	Entrepreneurship
0 7	No
1 6	No
2 6	No

2)List all numerical and categorical attributes.**Numerical Attributes**

- Age
- High_School_GPA
- SAT_Score
- University_GPA
- Internships_Completed
- Projects_Completed
- Certifications
- Soft_Skills_Score
- Networking_Score
- Job_Offers
- Starting_Salary
- Career_Satisfaction
- Years_to_Promotion
- Work_Life_Balance

Categorical Attributes

- Student_ID
- Gender
- Field_of_Study
- Current_Job_Level
- Entrepreneurship

3)Explain why normalization is required for the selected dataset.

The dataset contains attributes with significantly different scales.

Examples:

Feature	Range
University_GPA	2.0 – 4.0
SAT_Score	900 – 1600
Starting_Salary	25,000 - 150,000
Networking_Score	1 – 10

Without normalization, attributes with larger values such as SAT Score and Starting Salary may dominate the learning process of machine learning algorithms. Normalization ensures that all numerical features contribute equally during model training and improves convergence, accuracy, and interpretability.

Part B: Data Preparation

1) Load and inspect the dataset.

The dataset was loaded using Pandas and inspected using:

```
df.head()
```

```
df.info()
```

```
df.describe()
```

The dataset contains 400 records and 19 attributes.

2) Handle missing values (if any).

Missing values were checked using:

```
df.isnull().sum()
```

No missing values were found in the dataset.

3) Remove duplicate records (if any).

Duplicate records were checked using:

```
df.duplicated().sum()
```

No duplicate records were identified.

4) Visualization Before Normalization

The following visualizations were generated before normalization:

- Histogram of Starting Salary
- Boxplot of SAT Score
- Bar Plot of Average Salary by Field of Study
- Correlation Heatmap
- Line Plot showing relationships among selected variables

The plots revealed substantial differences in feature scales and the presence of varying distributions.

5) Visualization After Normalization

After applying Min-Max Normalization and Z-Score Standardization:

- Histograms showed transformed distributions.
- Boxplots displayed scaled feature ranges.
- Comparative plots illustrated reduced scale differences.
- Features became more comparable for machine learning purposes.

Part C: Normalization Techniques

1) Min-Max Normalization

Formula

Min-Max Normalization transforms data into a fixed range between 0 and 1.

Min-Max Normalized Value:

$$(X - X_{\min}) / (X_{\max} - X_{\min})$$

Applied Features

- SAT_Score
- University_GPA
- Starting_Salary
- Networking_Score
- Soft_Skills_Score

Observations

- All transformed values lie between 0 and 1.
- Relative relationships among observations remain unchanged.
- Suitable for algorithms that require bounded feature ranges.

2) Z-Score Normalization (Standardization)

Formula

$$Z = (X - \mu) / \sigma$$

where:

- μ = Mean
- σ = Standard Deviation

Applied Features

- SAT_Score
- University_GPA

- Starting_Salary
- Networking_Score
- Soft_Skills_Score

Observations

- Mean becomes approximately 0.
- Standard deviation becomes approximately 1.
- Outliers remain visible but feature scales become comparable.

Part D: Analysis

1) Compare the outputs obtained from different normalization techniques.

- The dataset contains 400 records and 19 attributes.
- No missing values or duplicate records were identified.
- Academic performance, internships, projects, and networking influence career outcomes.
- Career success is affected by both technical and interpersonal skills.
- Significant variation exists across different fields of study.

2) Identify which normalization method is most suitable for the selected domain and justify your answer.

For this Education and Career Success dataset, **Z-Score Normalization** is considered more suitable.

Justification

- The dataset contains attributes with large differences in scale.
- Features such as Starting Salary and SAT Score have wider ranges compared to GPA and Networking Score.
- Z-Score Standardization reduces scale differences while preserving statistical characteristics.
- It is highly effective for machine learning algorithms such as Logistic Regression, Support Vector Machines, K-Means Clustering, and Neural Networks.

Therefore, Z-Score Normalization provides a more balanced representation of student and career-related attributes and is recommended for subsequent machine learning tasks.

Conclusion

The Education and Career Success dataset was successfully preprocessed using Min-Max Normalization and Z-Score Standardization. Both methods effectively scaled the numerical attributes and improved feature comparability. Comparative analysis indicated that Z-Score Normalization is more suitable for this dataset due to the varying feature ranges and the presence of potentially influential attributes such as salary and SAT scores. The normalized dataset is now ready for machine learning applications such as employability prediction, career success analysis, salary prediction, and workforce readiness assessment.